

ISHA PATRO

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EDUCATION

NEW YORK UNIVERSITY
Master of Science in **Financial Engineering**
GPA: 3.6/4.0

Brooklyn, NY
Expected 05/27

VELLORE INSTITUTE OF TECHNOLOGY, AP
Bachelor in **Computer Science**
GPA: 3.5/4.0

Andhra Pradesh, India
07/2

PROGRAMMING & TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, R, VBA

Web Development: React, RESTful APIs, AG Grid, Streamlit, Dash, Tablue

Data Science: Statistical Modeling, Machine Learning, Deep Learning, LLMs, RAG, AI Agents, AI Inference

Databases & Cloud: SQL, Neo4j, q/kdb+, AWS, MongoDB, Microsoft Excel, Bloomberg Terminal

Quantitative Finance: Financial Modeling, Portfolio Management, Fixed Income, Time Series Forecasting, Prediction Markets

EXPERIENCE

J.P MORGAN CHASE & CO. | Commercial & Investment Bank

01/21 - 08/25

Software Developer Engineer - II, MBS Division - Fixed Income Pricing Team (Banglore, India)

- Led the development and post-launch ownership of Trade Activity Monitoring (TAM) as Epic Owner, collaborating with stakeholders to achieve a 50% latency reduction and significantly improve trading floor execution efficiency.
- Pioneered and architected a comprehensive Mortgage Matrix modernization initiative driving 12% latency improvement supporting \$5B+ annual trading volume.
- Designed and implemented Trade Reporting and Compliance Engine (TRACE) in an agile environment collaborating with cross-functional teams to boost data processing efficiency by 30% and streamline regulatory reporting workflows.

RESEARCH & OPEN SOURCE CONTRIBUTION

Temporal Graph Neural Additive Model for Portfolio Risk Forecasting

09/26 - Present

Master's Thesis

- Building a leakage-aware, interpretable graph-temporal machine learning pipeline for multi-equity volatility forecasting, portfolio risk estimation, and optimization research across U.S. equities with purged walk-forward validation.

Modeling Volatility in Polymarket Prices with Binary Jump-Tweedie Dynamics

07/26

Under review for the Prediction Markets Conference, Yale School of Management & Swensen Asset Management Institute

- Modeled market volatility using arcsine probability geometry, Tweedie likelihood estimation, market-level clustering, & category-specific parameter selection, benchmarking performance against GARCH and Hidden Markov baselines.

Quantitative Finance Corpus Architecture

07/26

- Collaborated with finance professionals to curate a corpus of fixed-income and derivatives problems, guiding AI agents to match Excel calculations, improving mathematical precision and reducing hallucinations in quantitative finance AI systems.

FinOKF - Clearbox Financial Research and Caching System

06/25 - 09/25

Under review at Journal of Association for the Advancement of Artificial Intelligence

- Developed an enterprise SEC filing intelligence platform that uses Google's Open Knowledge Format inspired RAG system and per-session LRU(Least Recently Used) caching to minimize token usage, latency, API costs for equity research.

Narrative Risk Allocation for Commodity Portfolios

01/26 - Present

- Engineering an end-to-end differentiable deep-learning portfolio allocator that embeds 125K+ financial news articles into volatility-spillover signals across 9 commodity futures, achieving a 1.73 Sharpe ratio over a 16-year walk-forward backtest.

Implied Prepayment Surfaces via Finance-Informed Neural Networks in Agency MBS Markets

11/25 - 03/26

Under review at Journal of Computational Finance.

- Built a finance-informed neural network that inverts agency TBA prices into a market-implied prepayment (CPR) surface under a Hull-White no-arbitrage constraint, showing the implied-realized divergence is a predictor of forward TBA returns

gutePandas (Open Source Contribution)

06/25 - Present

- Published a high-performance Pandas-compatible execution layer backed by kdb+/q, achieving order-of-magnitude speedups(10x-100x+) on large-scale analytical workloads while preserving Python-first research ergonomics.

Unlocking Financial Wisdom: Building Knowledge Graphs from Investopedia

06-24 - 08/24

Published in International Journal of Research and Analytical Reviews (IJRAR), Sep 2024

- Built a financial knowledge graph from 6,000+ Investopedia terms using data extraction and relationship mapping, enabling semantic visualization in Obsidian; presented at PitchWell Season 3 Finals at J.P. Morgan

HONORS / AWARDS

- Awarded the prestigious **Numerix Women in Finance Scholarship**, a **\$20,000 honor**, 2025
- Two-time 1st Place Winner, Engineering Clinics Expo, VIT (**\$1,200**) - across all engineering disciplines, 2017-201

ACADEMIC PROJECTS

2WIN

06/26

- Designed and backtested quantitative prediction-market strategies using tick-level order-book data, dynamic risk-based sizing, automated execution logic, and portfolio PnL attribution across real-time Polymarket markets.

VORTEX

05/26

- Architected a full-stack quantitative risk analytics platform: Gaussian HMM regime detection, 5-method VaR backtesting, XGBoost volatility forecasting, and a LangGraph multi-agent LLM system.

CryptoHedgeLab

03/26

- Engineered a high-performance crypto trading and backtesting system in C++ with kdb+/q, achieving microsecond-level execution, real-time hedging, and tick-level data storage for deterministic strategy evaluation.

Geometry of Intelligence

01/25 - Present

- Created an independent machine learning research series explaining algorithms from first principles through mathematical intuition, probability, and clean code implementations.

Cross-Currency Dynamics in Cryptocurrencies under Stablecoin Regulation

01/26 - 03/26

- Analyzed cross-currency basis and liquidity fragmentation in crypto markets using high-frequency data across exchanges, evaluating stablecoin-driven pricing inefficiencies and regulatory impact (GENIUS Act)

Monte Carlo Pricing of European Barrier Options

10/25

- Developed a Monte Carlo engine under risk-neutral GBM to price path-dependent barrier options (Down-and-Out, Knock-In/Out), incorporating barrier monitoring and sensitivity analysis.

Advanced NASDAQ Market Prediction System with Deep Learning Optimization

08/25

- Engineered a sophisticated LSTM neural network system for high-frequency NASDAQ market prediction, achieving 26.7% improvement in forecasting accuracy through advanced hyperparameter optimization.

Minted Wealth

03/25

- Developed a comprehensive financial intelligence dashboard integrating live market data, portfolio performance analytics, and net worth insights for smarter investment decisions.